Name: Mohammed Haris

Roll no: 261100690006

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Security Features in Real-World Applications

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

Sl No	Application	Security Feature	Security Mechanism	Product/Technology	How it provides Security
1	Signal	Confidentiality	Encryption/E2EE	Signal Protocol, Double Ratchet	Ensures only intended recipients can read message/call content
		Authentication	Identity verification	Safety Numbers	Helps detect man-in-the-middle attempts when verifying contacts/devices
		Integrity	Encryption is Authenticated	AEAD checks	Ciphertext tampering is detectable
		Authorisation	Secure device registration		makes it harder for attackers to add devices or hijack the account's cryptographic state
2	Amazon Store	Authentication	Password + MFA	MFA through Passkeys, TOTP and OTP	Reduces account takeover and fraudulent checkout

		Confidentiality	Encryption in transit, Payment protection	TLS/SSL, Session Tokens	Prevents interception of credentials, addresses, and checkout/session data
		Integrity	Session integrity		Blocks unauthorised actions that could change orders or payment methods
		Availability	Built-in high availability infrastructure	Fault tolerance ecommerce infra	If a customer gets stuck at a payment interface, refund or dispute mechanisms are built-in
		Authorisation	Account based access		
3	Zoom Video	Authentication	Password + MFA	SSO via SAML, OAuth	Enforces organization-wide sign-in policy for who can access Zoom features
		Authorisation	Meeting Passcodes + role-based permissions		Restricts who can join
		Confidentiality	Encryption	AES-256, TLS, optional E2EE	Protects meeting audio/video/screen share from interception.
		Integrity	Authenticated Meetings	Session Tokens	Enables mitigation if a participant behaves in a malicious manner.
4	Google Drive	Confidentiality	Encryption at rest and transit	Client-side encryption, SSL/TLS	Protects sign-in, file transfer and session data from interception

		Authenticatio n	Password + MFA	Passkeys, SSO using OpenID Connect, SAML 2.0, OAuth	Prevents login compromise from turning into file access
		Authorisatio n	RBAC and shared link access control		Limits who can access and edit what file or data
		Availability	Version history/restore options	Redundant replication	Maintains availability after file deletion by mistake
		Integrity	Audit logs		Supports investigations into unauthorised actions
5	Dropbox	Confidentialit y	Encryption at rest & transit	AES-256, E2EE/ Zero-knowledge & TLS/SSL	Prevents eves dropping
		Authenticatio n	Password + MFA	SSO using SAML/OAuth	Reduces account takeover if password is compromised
		Availability	Version history	Distributed storage	If storage is accidentally deleted, improves availability of data
		Authorisatio n	Access Control on shared links		Restricts sensitive admin actions to authorised admins
6	Bitwarden	Confidentialit y	Encryption at rest and transit	AES-256, TLS for vault syncs, Zero-knowledge infrastructure	Prevents reading passwords if vault storage/sync storage is accessed
		Integrity	Audit log + versioning		Protects against scenarios where account is still logged in but attacker uses unlocked session

		Authentication	Master password + MFA	PBKDF2	Prevents account take-over. PBKDF2 is used to produce strong fixed-length key
		Availability	Device sync + local access	Offline/online continuity	Local storage option available

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Sl No.	Application	Controls and Protocols	Prevention	Detection	Recovery
1	Signal	Confidentiality - E2EE via signal protocol	Prevents the service from reading message contents	Identity verification helps detect MITM during device verification	Secure session continues after reconnects through cryptographic ratcheting and device/session state management.
		Integrity via signal protocol	Undetected message modification	Failed authentication/decryption effectively detects tampering at the app layer	Messages which comes next remain secure after transient failures
2	Dropbox	Confidentiality using TLS, AES-256	Prevents eavesdropping and direct	Activity logs	Restores access to legitimate files via file

		encryption at rest	readout of stored files		restore or version history
		Integrity	Prevents unauthorised edits	Activity logs	Version history and rollback feature
		Availability	Prevents outages at server level	Operational monitoring	Restore deleted or overwritten files
3	Zoom Video	Confidentiality using TLS and E2EE feature for meetings	Prevents interception of meeting audio, video, screen share	Account and meeting security controls	Meetings can be often restarted
		Availability	Prevents meeting drop-off from completely failing	Telemetry at the server level	Clients can reconnect and re-establish meeting state
		Integrity	Prevents basic tampering of meeting video and audio	Meeting controls	Host can lock or allow only allowed clients and individuals
4	Bitwarden	Confidentiality using AES 256, PBKDF2 or Agon2 for key generation	Zero-knowledge design. Prevents the reading of the vault if storage is stolen or intercepted in transit	Login alerts, vault access event logging	Ability to resync vault after issues
		Integrity	Prevents attackers form altering encrypted vaults	Unexpected vault change events	Rederivation of keys allows consistent decryption if master password is correct

		Availability with multi device and local sync	Prevents single device failure form permanently losing access	Detects inability to unlock password manager	Account recovery options are limited due to keys being private and provider don’t have access to them
5	Amazon	Confidentiality using HTTPS/TLS while browsing	Prevents packet sniffing from reading credentials and other user details	Account telemetry and suspicious login patterns	Password resets and other recovery workflows
		Integrity	Prevents unauthorised users from modifying cart, price, order etc., details or performing fraudulent transactions	Risk detection that flags suspicious checkout behaviour	Order cancellation, refunds and customer support
		Availability	Prevents service failures since Amazon servers are robust	Alerts and monitoring on outages, payment failures etc.,	Dispute resolution during outages through customer support